

Supported zero on an evaluated arithmetic source plane

Split-Zero research programme

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1 The full endpoint plane and the supported-zero action

The original cosine endpoint test has a sine companion. This note evaluates their complete reflected Weil matrix and carries that matrix through the actual even-source prime receiver. Supported-zero multiplication then has a fully determined metric defect: its difference from the orthogonal projection onto its image is square-zero. The defect vanishes at one explicitly derived positive time. Its analytic family has a cohomology class supported at that time, lying over the supported-zero prime of the original arithmetic base.

There are two time parameters, which are never identified in the proof: the theta time is fixed at $t = 1/32$; $T > 0$ is the independent integer-coefficient heat time of the supported-zero receiver. The calculation at $T = 1/32$ is a stated specialization of the latter.

1.1 The original reflected form on both source directions

Keep the entire original density Φ , the original coordinates, and the function $H = H_{1/32}$ of AP2 in the complete appendix. Define

$$\begin{aligned} h_c(x) &= e^{-8x^2} H(x) \cos(16x), & h_s(x) &= e^{-8x^2} H(x) \sin(16x), \\ f_c &= (D^2 - 1/4)h_c, & f_s &= (D^2 - 1/4)h_s. \end{aligned} \tag{TS1}$$

The definitions of H and Φ , including both terms of every theta summand, are the original Rodgers–Tao formulas [phidef](#), [htdef](#), [hoz](#), [sas](#). Their original LaTeX and exact factors are used in AP2–3.

The Mellin convention remains $F_f(s) = \int f(x)e^{-(s-1/2)x} dx$. Put $w = s - 1/2$, $\sigma_1 = 1 + itw$, $\sigma_0 = itw$, and write out the two complete original-zeta terms

$$\begin{aligned} Z_1(s) &= e^{t(w-i/(2t))^2} \sigma_1(\sigma_1 - 1) \pi^{-\sigma_1/2} \Gamma(\sigma_1/2) \zeta(\sigma_1), \\ Z_0(s) &= e^{t(w+i/(2t))^2} \sigma_0(\sigma_0 - 1) \pi^{-\sigma_0/2} \Gamma(\sigma_0/2) \zeta(\sigma_0). \end{aligned} \tag{TS2}$$

Every exceptional product is continued as the displayed full product. The Gaussian integration in AP3 proves, with no discarded factors,

$$F_c(s) = \frac{s(s-1)\sqrt{\pi t}}{16} (Z_1 + Z_0), \quad F_s(s) = \frac{s(s-1)\sqrt{\pi t}}{16i} (Z_1 - Z_0). \tag{TS3}$$

The sources are real, f_c is even and f_s is odd. Each has Gaussian decay with all derivatives, and both transforms vanish at the two prescribed endpoints 0, 1. No assertion that these are their only zeros is made here.

On their complex span $V = \text{span}_{\mathbb{C}}\{f_c, f_s\}$, use the conjugate-linear-first form

$$\langle f, g \rangle_W = \sum_{\rho} m_{\rho} \overline{F_f(1 - \bar{\rho})} F_g(\rho). \tag{TS4}$$

The sums converge absolutely by the same Gaussian bounds and original zero count used in AP4–6. The reflected source is $f^\#(x) = \overline{f(-x)}$, so the complete source convolution for TS4 is $k_{f,g} = f^\# * g$. For any such convolution, the original arithmetic receiver is

$$\begin{aligned} P(k) &= \sum_{n \geq 2} \frac{\Lambda(n)}{\sqrt{n}} [k(\log n) + k(-\log n)], \\ A_\infty(k) &= -(\gamma_E + \log \pi)k(0) + \int_0^\infty \frac{e^{-x}k(0) - \frac{1}{2}e^{-x/4}[k(x/2) + k(-x/2)]}{1 - e^{-x}} dx, \\ \langle f, g \rangle_W &= A_\infty(k_{f,g}) - P(k_{f,g}). \end{aligned} \quad (\text{TS5})$$

Apply the finite-cutoff calculation AP6 to $A(s) = \overline{F_f(1 - \bar{s})}F_g(s)$. It retains every $A(-2m)$, the pole at one, and the matching Gamma boundary; both endpoint terms vanish for these actual tests. This proves TS5 without assigning a value to an individually divergent trivial-zero series. The source conventions are those of [Alain Connes, Appendix II, \(1\)–\(11\), Theorems 1 and 6](#); the Gaussian extension and all boundary terms are proved in the appendix.

For the cross term, $k_{c,s} = f_c * f_s$ is odd, and $k_{s,c} = -k_{c,s}$. Both prime legs cancel, the symmetric Gamma numerator is zero, and $k_{c,s}(0) = 0$. Hence

$$G = \begin{pmatrix} K_c & 0 \\ 0 & K_s \end{pmatrix}, \quad K_c = \langle f_c, f_c \rangle_W, \quad K_s = \langle f_s, f_s \rangle_W, \quad 50 < K_c, K_s < 64. \quad (\text{TS6})$$

This is the original basis, without rescaling either source.

Here are the additional calculations proving the second bound. If

$$U_H = (256x^2 - 1089/4)H - 32xH' + H'', \quad V_H = 512xH - 32H',$$

then

$$f_c = e^{-8x^2}(U_H \cos 16x + V_H \sin 16x), \quad f_s = e^{-8x^2}(U_H \sin 16x - V_H \cos 16x). \quad (\text{TS7})$$

Replacing H by the explicitly separated constant $C = J_0$ defines comparison sources; all residuals remain, as in AP7–18. The two sine/cosine formulas have exactly the same absolute envelopes A, E, B, E_1 . The odd autocorrelation is $k_s = -f_s * f_s$, not $f_s * f_s$. Its exact comparison is

$$\begin{aligned} k_{s,0}(x) &= \frac{C^2 \sqrt{\pi}}{128} e^{-4x^2} \{ (65536x^4 - 1622528x^2 + 1250369) \cos 16x \\ &\quad + 2048x(256x^2 - 1121) \sin 16x \\ &\quad - e^{-16}(65536x^4 - 49664x^2 + 3137) \}. \end{aligned} \quad (\text{TS8})$$

Indeed $-h_{s,0} * h_{s,0} = C^2 \sqrt{\pi} e^{-4x^2} (\cos 16x - e^{-16})/8$; apply $(D^2 - 1/4)^2$. The comparison differs from AP14 only in the displayed opposite-frequency term; that term remains in the finite computation.

The identical source bounds imply the identical residual estimates $d_0 < 0.303$, $d_2 < 175$, $E_\Gamma < 6.1$, $E_P < 0.45$. AP19–23 prove the full prime tail $< 10^{-21}$, exact Gamma tail contribution and residual $< 10^{-23}$, and the explicit origin-interval bound. The 100-bit interval calculation `certify_pair.py` evaluates TS8 and the original moments, with source-tail radius 10^{-300} proved in the appendix. For the sine comparison it gives

$$56.95 < A_\infty(k_{s,0}) - P_{s,0,64} < 56.97.$$

Adding the entire residual as in AP26 proves TS6. Thus for every $(a, b) \neq (0, 0)$,

$$50(|a|^2 + |b|^2) < \langle af_c + bf_s, af_c + bf_s \rangle_W < 64(|a|^2 + |b|^2). \quad (\text{TS9})$$

In the unscaled exponential basis $f_{\pm} = f_c \pm if_s$, the matrix is

$$G_{\pm} = \begin{pmatrix} K_c + K_s & K_c - K_s \\ K_c - K_s & K_c + K_s \end{pmatrix}. \quad (\text{TS10})$$

This follows by substituting $a = z_+ + z_-$, $b = i(z_+ - z_-)$; it retains the factor two in its eigenvalues $2K_c, 2K_s$.

1.2 The exact map into the original even prime source

The original moment receiver uses $H_{\text{ev}} = \mathcal{S}(\mathbb{R})^{\text{even}}$, with $m(h) = (h(0), \int h)$. The odd source f_s is therefore not an input to that receiver. The following exact arrow provides the connection, retaining its domain:

$$U : V \longrightarrow H_{\text{ev}}, \quad U(af_c + bf_s) = af_c + bf'_s. \quad (\text{TS11})$$

Set the actual constants

$$A = -J_2 - 1089C/4, \quad B = -48J_2 - 4868C, \quad R = -\frac{\sqrt{\pi/32}}{32}e^{-8}. \quad (\text{TS12})$$

Direct differentiation of TS7 gives $f'_s(0) = B$. Oddness gives $f_s(0) = \int f_s = 0$; decay gives $\int f'_s = 0$. AP27 gives the two moments of f_c . Consequently

$$mU(af_c + bf_s) = M \begin{pmatrix} a \\ b \end{pmatrix}, \quad M = \begin{pmatrix} A & B \\ R & 0 \end{pmatrix}, \quad M^{-1} = \begin{pmatrix} 0 & 1/R \\ 1/B & -A/(BR) \end{pmatrix}. \quad (\text{TS13})$$

All three constants are strictly negative; $\det M = -BR \neq 0$. This proves that TS11 is injective and provides its inverse on its image from the two original moments. Its construction is not asserted to preserve the form evaluated directly on Uf : every transported form below explicitly uses U^{-1} .

For completeness the algebraic prime receiver itself is as follows. Put $\mathcal{R} = \mathbb{C}[\mathbb{Q}_{>0}^{\times}]$, $I = \ker \text{aug}$, $M_0 = \ker(\text{aug} \otimes m)$, and $Q_0 = M_0/IM_0$. The moment lifts $(1 - 2\pi y^2)e^{-\pi y^2}$, $2\pi y^2 e^{-\pi y^2}$ have moments $(1, 0)$, $(0, 1)$. Hence

$$Q_0 \cong \ker m \oplus (I/I^2) \otimes \mathbb{C}^2, \quad I/I^2 \cong \bigoplus_p \mathbb{C}\ell_p. \quad (\text{TS14})$$

The identity $[t_{ab} - 1] = [t_a - 1] + [t_b - 1]$ modulo I^2 and the independent valuations v_p prove the second isomorphism. The first follows by writing $M_0 = (\mathcal{R} \otimes \ker m) \oplus (I \otimes \mathbb{C}^2)$ and multiplying by I . In particular put $\Phi(\sum_a t_a \otimes h_a) = \sum_a h_a$ and $\beta(\sum_a t_a \otimes h_a) = \sum_p \ell_p \otimes \sum_a v_p(a)m(h_a)$. Both descend to the displayed quotient: multiplication by $t_b - 1$ kills Φ , and its β value is $\sum_p \ell_p \otimes v_p(b)m(\Phi c) = 0$ for $c \in M_0$. Then

$$\mathcal{B}_p : V \longrightarrow Q_0, \quad \mathcal{B}_p f = [(t_p - 1) \otimes Uf], \quad (\Phi, \beta)\mathcal{B}_p f = (0, \ell_p \otimes mUf) \quad (\text{TS15})$$

is an isomorphism onto the complete two-dimensional boundary summand at the original prime p . Its inverse is TS13, followed by $(a, b) \mapsto af_c + bf_s$. These are the same quotient maps as [FR15–25](#); the displayed proof fixes their action on this new source.

The transported positive form on this summand, in the unchanged moment coordinates (m_0, m_1) , is exactly

$$Q_B(m_0, m_1) = K_c |m_1/R|^2 + K_s \left| \frac{m_0 - Am_1/R}{B} \right|^2. \quad (\text{TS16})$$

This uses the specified inverse of TS15. The original periodization seminorm of this boundary summand remains zero because $\Phi = 0$. Equation TS16 does not identify that seminorm with a positive norm; it gives the explicit form brought from TS4 by the injective receiver.

1.3 Supported zero gives a specified oblique projection

Retain the original full bounded distributive lattice L and $G_L(\mathbb{Z})$, with $e = (0, 1_L)$, $\tau = (0, 0_L)$. Multiplication by e sends each supported integer to e and fixes every zero label z_λ . On finite supported-integer coefficients it is therefore $a \mapsto (\sum_n a_n)\delta_0$. Under the actual coefficient heat map

$$A_T a(y) = \sum_{n \in \mathbb{Z}} a_n \frac{e^{-(y-n)^2/(4T)}}{\sqrt{4\pi T}}, \quad T > 0,$$

the two moments are $(\sum_n a_n k_T(n), \sum_n a_n)$, where $k_T(n) = (4\pi T)^{-1/2} e^{-n^2/(4T)}$. After multiplication by e they are $(\kappa_T m_1, m_1)$, $\kappa_T = (4\pi T)^{-1/2}$. Every moment pair occurs already for an even coefficient vector:

$$a_0 = \frac{m_0 - k_T(1)m_1}{k_T(0) - k_T(1)}, \quad a_1 = a_{-1} = \frac{k_T(0)m_1 - m_0}{2[k_T(0) - k_T(1)]}. \quad (\text{TS17})$$

All other coefficients are zero. The denominator is positive; substitution proves both moments. Thus the induced endomorphism of the actual prime-boundary summand is well-defined, independently of the chosen coefficient lift, and is

$$J_T = \begin{pmatrix} 0 & \kappa_T \\ 0 & 1 \end{pmatrix}. \quad (\text{TS18})$$

No multiplication on the whole vector-space quotient is postulated. On the source plane the exact conjugate is

$$E_T = M^{-1} J_T M = \begin{pmatrix} 1 & 0 \\ c(T) & 0 \end{pmatrix}, \quad c(T) = \frac{\kappa_T R - A}{B}, \quad E_T^2 = E_T. \quad (\text{TS19})$$

The range is $\mathbb{C}u_T$, $u_T = (1, c(T))^T$, and the kernel is $\mathbb{C}(0, 1)^T$. It acts as the identity on its range and is not an invertible operator on the entire plane. The unital algebra map $\mathbb{C}[q]/(q^2 - q) \rightarrow \text{End}(V)$, $q \mapsto E_T$, is faithful, since the range and kernel are both nonzero. It factors through neither $q \mapsto 0$ nor $q \mapsto 1$. This states exactly how the supported-zero datum survives in the actual linear representation.

For the positive metric G of TS6,

$$GE_T - E_T^* G = \begin{pmatrix} 0 & -K_s c(T) \\ K_s c(T) & 0 \end{pmatrix}, \quad \|E_T\|_G^2 = 1 + \frac{K_s}{K_c} c(T)^2. \quad (\text{TS20})$$

The norm formula follows by maximizing $(K_c + K_s c^2)|a|^2/(K_c|a|^2 + K_s|b|^2)$, with equality at $b = 0$. Since $A, R < 0$, there is exactly one positive time at which this projection is self-adjoint:

$$\boxed{T_* = \frac{(R/A)^2}{4\pi}, \quad 2.99 \cdot 10^{-15} < T_* < 3 \cdot 10^{-15}.} \quad (\text{TS21})$$

The unique equation is $\kappa_{T_*} = A/R > 0$; the interval is certified from the original moments. At the stated specialization $T = 1/32$, the same certificate gives

$$-0.055923 < c(T) < -0.055922, \quad 1.00248 < \|E_T\|_G^2 < 1.00394. \quad (\text{TS22})$$

This is the actual transported action; no orthogonal projection has been substituted for it.

1.4 The square-zero metric defect and its exact algebra

For real $T > 0$, the G -orthogonal projection onto the same range is computed, with all metric factors, by

$$D_T = K_c + K_s c(T)^2, \quad P_T = \frac{1}{D_T} \begin{pmatrix} K_c & K_s c(T) \\ K_c c(T) & K_s c(T)^2 \end{pmatrix}. \quad (\text{TS23})$$

Indeed $P_T v = u_T(u_T^* G v)/D_T$, so its range, identity on that range, idempotence and self-adjointness follow directly. Define the exact difference

$$N_T = E_T - P_T = \frac{K_s c(T)}{D_T} \begin{pmatrix} c(T) & -1 \\ c(T)^2 & -c(T) \end{pmatrix}. \quad (\text{TS24})$$

Because $E_T P_T = P_T$ and $P_T E_T = E_T$, $N_T^2 = 0$. For the unchanged frame

$$\mathcal{F}_T = [u_T, w_T], \quad w_T = (-K_s c(T), K_c)^T, \quad \det \mathcal{F}_T = D_T,$$

one has

$$\mathcal{F}_T^{-1} N_T \mathcal{F}_T = \begin{pmatrix} 0 & -K_s c(T) \\ 0 & 0 \end{pmatrix}. \quad (\text{TS25})$$

This proves $N_T \neq 0$ precisely for $T \neq T_*$. Moreover u_T and w_T are G -orthogonal, with $\|u_T\|_G^2 = D_T$, $\|w_T\|_G^2 = K_c K_s D_T$. TS25 therefore proves the exact identity

$$\|N_T\|_G^2 = (K_s/K_c) c(T)^2, \quad \|E_T\|_G^2 = 1 + \|N_T\|_G^2. \quad (\text{TS25a})$$

For each such T , the map

$$\mathbb{C}[\eta]/(\eta^2) \longrightarrow \mathbb{C}[I, N_T] \subset \text{End}(V), \quad a + b\eta \longmapsto aI + bN_T \quad (\text{TS26})$$

is an isomorphism: multiplicativity follows from $N_T^2 = 0$, and injectivity follows from trace and $N_T \neq 0$. At T_* its evaluated image is $\mathbb{C}I$ and its kernel is exactly (η) . Thus the nilpotent is a computed defect of this original supported-zero action and its actual metric. It is not identified with the idempotent e itself.

1.5 Cohomology and representation specialization

Take the complex disc $\Omega = \{|T - T_*| < T_*/2\}$, and the branch of $\sqrt{T}$ positive at T_* . Retain TS19, TS23 and TS24 as holomorphic formulas on this disc. TS23 is the analytic continuation of the real orthogonal-projection formula; it is not claimed to be a Hermitian projection at nonreal T .

These formulas have no hidden denominator zero. In fact

$$0 < A/B < 1/16,$$

since $16(J_2 + 1089C/4) < 48J_2 + 4868C$, and $c(T) = (A/B)[(T_*/T)^{1/2} - 1]$. On Ω , $|c(T)| < (1 + \sqrt{2})/16 < 1/4$, whence $|K_s c(T)^2| < 4 < K_c$. Thus $D_T \neq 0$, and the frame TS25 is holomorphic and invertible on the whole disc.

Let $\mathcal{O}$ be the ring of holomorphic germs at T_* . The coefficient $n(T) = -K_s c(T)$ has the complete factorization

$$\begin{aligned} c(T) &= -\frac{R(T - T_*)}{B\sqrt{4\pi}\sqrt{T}\sqrt{T_*}(\sqrt{T} + \sqrt{T_*})}, \\ n(T) &= \frac{K_s R(T - T_*)}{B\sqrt{4\pi}\sqrt{T}\sqrt{T_*}(\sqrt{T} + \sqrt{T_*})}. \end{aligned} \quad (\text{TS27})$$

The displayed unit and its sign are retained; in particular $n'(T_*) = K_s R / [2B\sqrt{4\pi}T_*^{3/2}] > 0$. The actual two-periodic complex with every term $\mathcal{O}^2$ and differential N_T has

$$H^j \cong \mathcal{O}/(n(T)) \cong \mathcal{O}/(T - T_*) \quad \text{for every integer } j. \quad (\text{TS28})$$

To prove it, use the full frame TS25. An element $au_T + bw_T$ is in the kernel exactly when $n(T)b = 0$. The germ ring is a domain and $n \neq 0$, so the kernel is $\mathcal{O}u_T$; the image is $n\mathcal{O}u_T$. Evaluation at T_* gives the second isomorphism, whose kernel is exactly the principal ideal proved in TS27. At any real $T \neq T_*$ the specialized complex is acyclic; at T_* its differential is zero and its cohomology is $\mathbb{C}^2$ in every degree. These assertions compute specialization directly, without assuming cohomology commutes with it. The additional fibre class is explicit. The natural map from $H^j \otimes_{\mathcal{O}} \mathbb{C}$ gives the line $\mathbb{C}u_{T_*}$ in fibre cohomology, while the other class is represented by w_{T_*} . The free resolution $0 \rightarrow \mathcal{O} \xrightarrow{n} \mathcal{O} \rightarrow \mathcal{O}/(n) \rightarrow 0$ shows that $\text{Tor}_1^{\mathcal{O}}(\mathcal{O}/(n), \mathbb{C}) = \mathbb{C}$. Thus the base-change sequence here is exactly

$$0 \longrightarrow \mathbb{C}u_{T_*} \longrightarrow H^j(V_{T_*}, N_{T_*}) \longrightarrow \mathbb{C}[w_{T_*}] \longrightarrow 0. \quad (\text{TS28a})$$

All three spaces and maps have been evaluated in the original frame. The quotient map sends $au_{T_*} + bw_{T_*}$ to b times the Tor generator in the displayed resolution by n . If instead the connecting map uses the coordinate $T - T_*$, its value is $bn'(T_*)$; the complete positive coefficient is the one given immediately after TS27.

There is also an exact algebra specialization. The map

$$\mathcal{O}[\eta]/(\eta^2) \longrightarrow \text{End}_{\mathcal{O}}(\mathcal{O}^2), \quad \eta \longmapsto N_T \quad (\text{TS29})$$

is injective. Trace kills a scalar coefficient of a relation $aI + bN = 0$; a nonzero entry of N then gives $b = 0$ in the domain $\mathcal{O}$. Its image is a free rank-two algebra. In the frame TS25 the representation of its two basis elements has columns $(1, 0, 0, 1)^T$ and $(0, n, 0, 0)^T$. Its cokernel is $\mathcal{O}^2 \oplus \mathcal{O}/(n)$: subtract the first row from the fourth and read the four coordinates. All two-column minors generate exactly $(n) = (T - T_*)$. The abstract algebra fibre at T_* is still the dual-number algebra; its evaluated representation has kernel (η) . This is the exact failure of injectivity under that specialization.

1.6 Where this class lies on the full split prime spectrum

The preceding local object has an exact map to the original arithmetic base. The source construction is the [original globalization note](#), [cor:prime-ideals](#) and [thm:zero-prime-not-irr](#), and its full lattice extension is [version 11](#), [def:lattice-split](#) and [thm:lattice-prime-classification](#). The needed statements are rederived here. For a ring R and the unchanged bounded distributive lattice L with $0_L \neq 1_L$, define

$$G_L(R) = \{(r, 1_L) : r \in R\} \cup \{z_\lambda = (0, \lambda) : \lambda \neq 1_L\}. \quad (\text{TS30})$$

Addition uses the ring sum and joins of labels, multiplication uses the ring product and meets. Nonzero amplitudes always have label 1_L ; cancellations remain $e = z_{1_L}$, and $\tau = z_{0_L}$. For a domain R , the ideal

$$P_e^R = \{z_\lambda : \lambda \in L\}$$

is prime, since a product has zero amplitude exactly when one factor has zero amplitude. It is the principal ideal (e) : $ez_\lambda = z_\lambda$ and every product by e has zero amplitude. For a ring prime $\mathfrak{p}$, let $P_{\mathfrak{p}}^R$ be the inverse image of $\mathfrak{p}$ under amplitude. These operations stay in the displayed carrier: any nonzero sum has a top join, and any nonzero product has two top input labels. The ring and lattice identities prove associativity, distributivity, additive identity τ , and multiplicative identity $(1, 1_L)$ coordinate by coordinate. A semiring ideal contains τ , is closed under addition,

and absorbs multiplication by every element; primality means that membership of a product forces membership of a factor.

For completeness all remaining prime points retain the full lattice. A prime ideal excluding e contains no top-supported ring element, since multiplying such an element by -1 and adding would put e in that ideal. It is therefore $Q_J = \{z_\lambda : \lambda \in J\}$, where J is a proper lattice ideal prime for meets. Conversely every such Q_J is prime: if a product has label $\lambda \wedge \mu \in J$, meet primality puts λ or μ in J . That label is not 1_L , so the corresponding factor is the zero-labelled element of Q_J . This argument also covers rings with zero divisors. A prime containing e contains every zero label and is exactly $P_{\mathfrak{p}}^R$ for a ring prime; the ring ideal and its primality follow from the original top-supported ring operations. This proves the classification with no quotient of L .

The germ ring $\mathcal{O}$ has precisely the ring primes (0) , $\mathfrak{m} = (T - T_*)$: every nonzero germ factors as a finite power of $T - T_*$ times its nonvanishing holomorphic unit. Hence

$$Q_J \subsetneq P_e^{\mathcal{O}} \subsetneq P_{\mathfrak{m}}^{\mathcal{O}}, \quad \text{Ann}_{\mathcal{O}} H^j = (T - T_*). \quad (\text{TS31})$$

The first symbols range over the actual lattice prime ideals; they are not all replaced by a single point. For the original two-element lattice only, the chain is $(\tau) \subsetneq (e) \subsetneq P_{\mathfrak{m}}$.

The original integer constants give a unital semiring map $G_L(\mathbb{Z}) \rightarrow G_L(\mathcal{O})$, preserving every label. Each nonzero integer is a unit in $\mathcal{O}$. Its map on prime spectra therefore satisfies exactly

$$Q_J \mapsto Q_J, \quad P_e^{\mathcal{O}} \mapsto P_e^{\mathbb{Z}}, \quad P_{\mathfrak{m}}^{\mathcal{O}} \mapsto P_e^{\mathbb{Z}}. \quad (\text{TS32})$$

The cohomology module TS28, considered by restriction along amplitude, has support $P_{\mathfrak{m}}^{\mathcal{O}}$ and thus lies over the supported-zero prime $P_e^{\mathbb{Z}}$. To prove the support directly, localization at Q_J inverts the annihilating scalar e ; localization at $P_e^{\mathcal{O}}$ inverts the annihilating nonzero germ $T - T_*$. Both modules therefore vanish. At $P_{\mathfrak{m}}^{\mathcal{O}}$, every inverted scalar has unit germ amplitude, so localization leaves $\mathcal{O}/(n)$ unchanged. No ordinary positive integer prime lies in the image of this complex-germ chart: each is a unit here. This specifies the chart and its arithmetic location; it does not identify its new specialization point with an ordinary integer prime or with the absent zero.

Restriction of the module to the integer base has a different support calculation, which is also explicit. Its underlying module is $\mathbb{C}$ with integer scalars. At every lattice prime Q_J , the element e belongs to the localization set and acts as zero, so the localized module vanishes. At $P_e^{\mathbb{Z}}$, and at every $P_{(p)}^{\mathbb{Z}}$, every element in the localization set acts by a nonzero integer and thus bijectively on $\mathbb{C}$. The localized module is consequently $\mathbb{C}$, not zero. Therefore

$$\text{Supp}_{G_L(\mathbb{Z})} \text{Res } H^j = V(e) = \{P_e^{\mathbb{Z}}\} \cup \{P_{(p)}^{\mathbb{Z}} : p \text{ a positive integer prime}\}. \quad (\text{TS32a})$$

This is the specialization closure of the chart image of the cohomology support. It retains every ordinary arithmetic prime. It is not obtained by treating a complex germ chart as a surjection onto the integer spectrum.

The module's scalar action via amplitude and the original monoid action TS19 are different explicitly related representations: the former sends e to zero, whereas the faithful idempotent algebra representation in TS19 sends it to $E_T \neq 0$. The latter cannot factor through the former's scalar map. Indeed every scalar action of $G_L(\mathcal{O})$ on an additive group kills each z_λ : $z_\lambda + z_\lambda = z_\lambda$ and cancellation in that group give $z_\lambda v = 0$. This proves exactly why the original nonzero E_T is a representation of the multiplicative monoid and its stated idempotent algebra, rather than such an additive scalar action. The label-preserving lift of TS26 sends η^2 to the supported zero endomorphism, not to τ ; thus even the square-zero calculation retains the original distinction. The dual-number algebra here is the proved defect algebra of the original supported-zero operator, not a replacement for its idempotent multiplication law.

1.7 Actual transport and the remaining zero-time boundary

The original family TS19 has the idempotent-preserving connection matrix

$$[E'_T, E_T] = E'_T = \begin{pmatrix} 0 & 0 \\ c'(T) & 0 \end{pmatrix}, \quad L(T, T_0) = \begin{pmatrix} 1 & 0 \\ c(T) - c(T_0) & 1 \end{pmatrix}. \quad (\text{TS33})$$

Direct multiplication proves the differential equation $\partial_T L = [E'_T, E_T]L$ and initial value $L(T_0, T_0) = I$. It also proves the intertwining identity $E_T L = L E_{T_0}$ and the composition identity $L(T_2, T_1)L(T_1, T_0) = L(T_2, T_0)$. This solves the stated commutator equation directly from the actual idempotent family. No uniqueness among all connections preserving its range is asserted. It is regular at T_* , despite the cohomology specialization there. It depends only on endpoints on the stated branch; the computed transport has trivial holonomy around loops in Ω .

The original zero-time boundary remains a different, quantitative question, for which the exact growth is already determined:

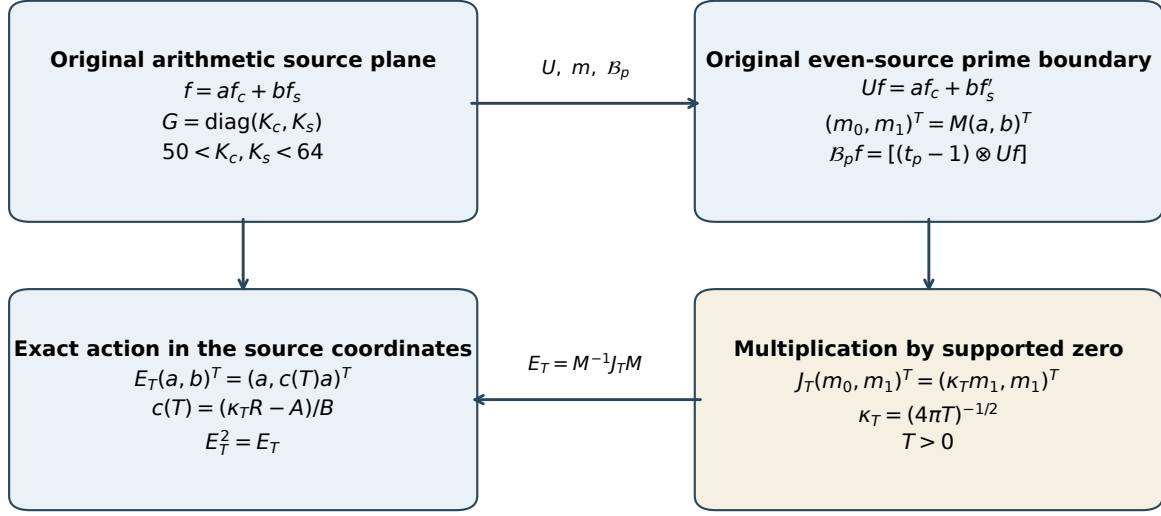
$$\|E_T\|_G^2 = 1 + \frac{K_s}{K_c B^2} \left[\frac{R^2}{4\pi T} - \frac{2RA}{\sqrt{4\pi T}} + A^2 \right], \quad T > 0. \quad (\text{TS34})$$

Every term is retained. The leading coefficient is strictly positive, so $\|E_T\|_G$ grows as a specified positive constant times $T^{-1/2}$; TS33 has the exact shear coefficient $(R/B)[(4\pi T)^{-1/2} - (4\pi T_0)^{-1/2}]$. These are properties of the original integer-coefficient heat receiver and its Weil metric on this source plane. They do not identify that receiver with the zero-moving de Bruijn–Newman flow. TS1–5 give their exact connecting tests and original-zeta formulas.

The result is a positive two-dimensional arithmetic form together with a non-self-adjoint supported-zero action, an exact nilpotent defect, and a cohomology class over the supported-zero prime. All four statements hold simultaneously. None assigns a sign to the Weil form on its full infinite-dimensional source space.

The evaluated maps and their prime support

The actual supported-zero action and its computed metric defect



P_T : the G -orthogonal projection onto $\text{im}(E_T)$

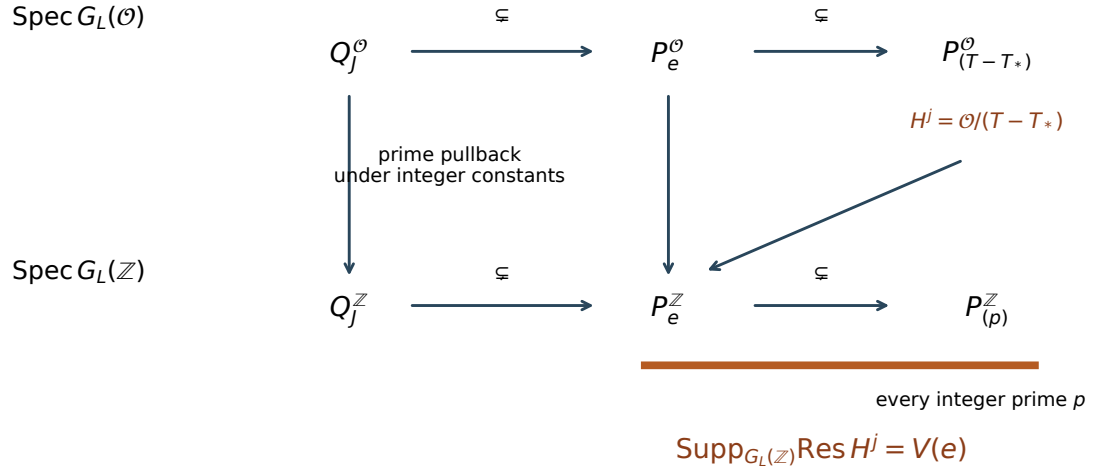
$$N_T = E_T - P_T, \quad N_T^2 = 0, \quad \|N_T\|_G^2 = (K_s/K_c)c(T)^2$$

$N_T = 0$ exactly at $T_* = (R/A)^2/(4\pi)$; theta time stays $t = 1/32$.

The upper arrow is TS11–15, with its exact inverse TS13. The original integer-coefficient action is TS17–19; the nilpotent is its metric defect TS23–26. The two independent times are labelled. The original density and trace conventions are those of Rodgers–Tao and Connes cited in TS1 and TS5, with every original-zeta factor in TS2.

The local chart and the retained arithmetic primes

The local geometric point and the full arithmetic module support



J ranges over all proper meet-prime ideals of the original lattice L .

Horizontal arrows: prime inclusions. Downward arrows: the actual spectrum map.

The restricted module retains all ordinary primes, even though the complex chart misses them.

TS27–32a proves every displayed map and support statement. The orange segment records restricted-module support, not a new spectrum arrow. No lattice prime has been removed. The original split base is defined in TS30, retaining the supported-zero prime and its ordinary-prime specializations.

Retained complete scalar proof and source-tail certificate

The following AP1–28 source is reproduced unchanged from the preceding accepted calculation. It supplies the full original source, contour receiver, infinite-tail proofs and residual estimates used in TS6–9. The preceding standalone edition is preserved.

A The actual endpoint test has a strictly positive arithmetic pairing

This calculation evaluates the test constructed by the endpoint-resonance calculation. It uses the original meromorphic Riemann zeta function, its prime powers, its pole, every trivial zero at each finite contour cutoff, and the full Gamma boundary. At the original time $t = 1/32$, the result is the certified rational enclosure

$$\boxed{50 < K_{1/32} < 64.} \quad (\text{AP1})$$

Here K_t is defined below for this one explicit test. The two negative source moments of the same test therefore do not give a negative Weil pairing. The calculation determines this test's sign; it does not assert positivity for every test.

A.1 Original source, coordinates, and zeta factors

Throughout, x, u are the original real coordinates and $t = 1/32$. Retain

$$\begin{aligned} \Phi(u) &= \sum_{n=1}^{\infty} (2\pi^2 n^4 e^{9u} - 3\pi n^2 e^{5u}) e^{-\pi n^2 e^{4u}}, \\ H(x) &= \int_0^{\infty} e^{u^2/32} \Phi(u) \cos(xu) du, \\ h(x) &= e^{-8x^2} H(x) \cos(16x), \quad f(x) = h''(x) - \frac{1}{4}h(x), \quad k = f * f, \\ F(s) &= \int_{\mathbb{R}} f(x) e^{-(s-1/2)x} dx. \end{aligned} \quad (\text{AP2})$$

The density and heat convention are those of Brad Rodgers and Terence Tao, *The de Bruijn–Newman constant is non-negative*, [version 5](#), original TeX equations `phidef`, `htdef`, `hoz`, and `sas`. Only those source formulas are used here; the present sign calculation uses no assumption about the location of zeta zeros.

For clarity, the test's exact return to the original function is displayed with every factor. Set $w = s - 1/2$, $\sigma_0 = itw$, $\sigma_1 = 1 + itw$. Then

$$\begin{aligned} F(s) &= \frac{s(s-1)\sqrt{\pi t}}{16} \left\{ e^{t(w-i/(2t))^2} \sigma_1(\sigma_1-1)\pi^{-\sigma_1/2}\Gamma(\sigma_1/2)\zeta(\sigma_1) \right. \\ &\quad \left. + e^{t(w+i/(2t))^2} \sigma_0(\sigma_0-1)\pi^{-\sigma_0/2}\Gamma(\sigma_0/2)\zeta(\sigma_0) \right\}. \end{aligned} \quad (\text{AP3})$$

Every product at an exceptional argument is the analytic continuation of the displayed full product. In particular the products at arguments zero and one both equal one. The pole of ζ at one and the pole of $\Gamma(s/2)$ at zero are included in this statement. To prove AP3, Gaussian integration gives

$$\int_{\mathbb{R}} e^{-x^2/(4t)} H_t(x) e^{-wx} dx = 2\sqrt{\pi t} e^{tw^2} H_0(2tw).$$

The two exponential terms of $\cos(x/(2t))$, followed by $16H_0(z) = s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$ with $s = 1/2 + iz/2$, give AP3. Absolute interchange follows from the outer Gaussian and the super-exponential original density. Two integrations by parts give the retained factor $s(s-1)$. Consequently $F(0) = F(1) = 0$, exactly.

A.2 Complete arithmetic receiver

Write $j = \zeta'/\zeta$, and let ρ run through the nontrivial zeros of the original function with multiplicity m_ρ . The exact quantity evaluated is

$$K = \sum_{\rho} m_{\rho} F(\rho)^2 = A_{\infty}(k) - P(k), \quad P(k) = 2 \sum_{n \geq 2} \frac{\Lambda(n)}{\sqrt{n}} k(\log n), \quad (\text{AP4})$$

where $\Lambda(p^a) = \log p$ for every prime power and $\Lambda(n) = 0$ otherwise. The full archimedean term is

$$A_{\infty}(k) = -(\gamma_E + \log \pi)k(0) + \int_0^{\infty} \frac{e^{-x}k(0) - e^{-x/4}k(x/2)}{1 - e^{-x}} dx. \quad (\text{AP5})$$

Evenness and reality give $\overline{F(1 - \bar{s})} = F(s)$. Thus AP4 has the actual reflected Weil product; away from the critical line it has not been replaced by an absolute square.

We recall the finite-cutoff derivation to retain the original divisor. Put $A = F^2$, $b_N = -2N - 1$, $U_N = \sum_{m=1}^N A(-2m)$, and $\kappa(s) = -\frac{1}{2} \log \pi + \frac{1}{2} \psi_{\Gamma}(s/2)$. For upward vertical integrals $I_c = (2\pi i)^{-1} \int_{\Re s=c}$, the argument principle and the full original functional equation give

$$\begin{aligned} V_N &= K + U_N - A(1) = I_{\sigma}(Aj) - I_{b_N}(Aj), \quad \sigma > 1, \\ j(s) + j(1-s) &= -\kappa(s) - \kappa(1-s), \\ V_N &= -P(k) + \mathcal{G}_N, \\ \mathcal{G}_N &= I_{b_N}\{A(s)[\kappa(s) + \kappa(1-s)]\} = A_{\infty}(k) + A(0) + U_N. \end{aligned} \quad (\text{AP6})$$

The rational derivatives of $s(s-1)$ in the original multiplier cancel with their reflected partners in the displayed reflection identity; the pole at one remains explicitly in V_N . On the right edge $j(s) = -\sum \Lambda(n)n^{-s}$; Mellin inversion gives the two prime legs. Moving the Gamma edge to $1/2$ crosses exactly the residues at $0, -2, \dots, -2N$, giving $A(0) + U_N$. The digamma integral on $1/2$ gives AP5 with its combined numerator. This proves AP4 because both endpoints vanish. No limit or finite value is assigned to U_N individually.

For the classical contour and measure conventions, see Alain Connes, *Trace formula in noncommutative geometry and the zeros of the Riemann zeta function*, [Appendix II, \(1\)–\(11\)](#), [Theorems 1 and 6](#). Its compact-support theorem is extended here by the actual Gaussian bounds below. They give arbitrary inverse-power decay of F on fixed vertical strips by integration by parts. The original zero count $O(T \log(T+2))$ gives absolute convergence of AP4, and contours at heights avoiding zero ordinates have vanishing horizontal integrals. All strips used in AP6 have finite width.

A.3 An exact decomposition with the entire residual retained

Define the positive original moments

$$J_j = \int_0^{\infty} u^j e^{u^2/32} \Phi(u) du, \quad C = J_0, \quad R(x) = H(x) - C. \quad (\text{AP7})$$

Every summand of Φ is positive for $u \geq 0$. The inequalities $e^{4u} \geq 1 + 4u + 8u^2$, $\pi > 3$, $\pi^2 < 10$ imply

$$J_j \leq 20j! \sum_{n \geq 1} \frac{n^4 e^{-3n^2}}{(12n^2 - 9)^{j+1}} < \frac{j!}{3^{j+1}}. \quad (\text{AP8})$$

Indeed $12n^2 - 9 \geq 3n^2$, and $\sum n^2 e^{-3n^2} < 1/20$. One fully rational check of the last inequality uses $e^3 > 2007/100$, $q = 1/8000$, and

$$\sum n^2 e^{-3n^2} < \frac{100}{2007} + \frac{1}{20} \left(\frac{1+q}{(1-q)^3} - 1 \right) < \frac{1}{20}.$$

Differentiation under the integral is therefore valid. In particular

$$|R(x)| \leq J_2 x^2/2, \quad |R'(x)| \leq J_2 |x|, \quad |R''(x)| \leq J_2, \quad |R'''(x)| \leq J_3. \quad (\text{AP9})$$

Put $f = f_0 + e_f$, where the equality, rather than a replacement of f , defines the residual:

$$f_0(x) = C e^{-8x^2} [(256x^2 - 1089/4) \cos(16x) + 512x \sin(16x)]. \quad (\text{AP10})$$

Direct differentiation of AP2 gives

$$e_f(x) = e^{-8x^2} \{ [(256x^2 - 1089/4)R - 32xR' + R''] \cos(16x) + [512xR - 32R'] \sin(16x) \}. \quad (\text{AP11})$$

No term of this residual is dropped from the pairing. With $r \geq 0$, define

$$\begin{aligned} A(r) &= 1089/4 + 512r + 256r^2, \\ E(r) &= 1 + 32r + (1345/8)r^2 + 256r^3 + 128r^4, \\ B(r) &= 4868 + 13060r + 12288r^2 + 4096r^3, \\ E_1(r) &= 48 + (3457/4)r + 3970r^2 + 7298r^3 + 6144r^4 + 2048r^5. \end{aligned} \quad (\text{AP12})$$

AP9–11 and their derivatives prove

$$\begin{aligned} |f_0(x)| &\leq C e^{-8x^2} A(|x|), \quad |e_f(x)| \leq J_2 e^{-8x^2} E(|x|), \\ |f'_0(x)| &\leq C e^{-8x^2} B(|x|), \quad |e'_f(x)| \leq e^{-8x^2} [J_2 E_1(|x|) + J_3]. \end{aligned} \quad (\text{AP13})$$

For an explicit derivative check, the cosine coefficient in $e^{8x^2} e'_f$ is $(13060x - 4096x^3)R + (768x^2 - 3265/4)R' - 48xR'' + R'''$; its sine coefficient is $(4868 - 12288x^2)R + 1536xR' - 48R''$. This proves every coefficient in AP12 without omitting the derivative of the original source.

A.4 Exact Gaussian convolution and uniform residual bounds

Write $k_0 = f_0 * f_0$ and $\Delta = k - k_0$. The exact convolution is

$$\begin{aligned} k_0(x) &= \frac{C^2 \sqrt{\pi}}{128} e^{-4x^2} \{ (65536x^4 - 1622528x^2 + 1250369) \cos(16x) \\ &\quad + 2048x(256x^2 - 1121) \sin(16x) \\ &\quad + e^{-16} (65536x^4 - 49664x^2 + 3137) \}. \end{aligned} \quad (\text{AP14})$$

To verify it, put $h_0(x) = C e^{-8x^2} \cos(16x)$. In the convolution set $y = x/2 + z$ and use the full product identity $\cos(16y) \cos(16(x-y)) = [\cos(16x) + \cos(32z)]/2$. The Gaussian integrals then give exactly

$$(h_0 * h_0)(x) = \frac{C^2 \sqrt{\pi}}{8} e^{-4x^2} [\cos(16x) + e^{-16}].$$

Since $f_0 = (D^2 - 1/4)h_0$, differentiation under the convolution gives $k_0 = (D^2 - 1/4)^2(h_0 * h_0)$. Expanding these two derivatives yields all coefficients in AP14, including the opposite-frequency contribution e^{-16} .

For any polynomial P with nonnegative coefficients put

$$\mathcal{I}(P) = \int_{\mathbb{R}} e^{-16x^2} P(|x|) dx, \quad \mathcal{I}(r^j) = 16^{-(j+1)/2} \Gamma((j+1)/2). \quad (\text{AP15})$$

The substitution $v = 16x^2$ on the two half-lines proves this identity with all constants. Define the explicit upper bounds

$$\begin{aligned} n_0 &= C\sqrt{\mathcal{I}(A^2)}, & n_e &= J_2\sqrt{\mathcal{I}(E^2)}, \\ n_1 &= C\sqrt{\mathcal{I}(B^2)}, & n_{e1} &= \sqrt{J_2^2\mathcal{I}(E_1^2) + 2J_2J_3\mathcal{I}(E_1) + J_3^2\mathcal{I}(1)}, \\ d_0 &= 2n_0n_e + n_e^2, & d_2 &= 2n_1n_{e1} + n_{e1}^2. \end{aligned} \quad (\text{AP16})$$

Since $\Delta = 2f_0 * e_f + e_f * e_f$, Cauchy–Schwarz and $k'' = f' * f'$ give

$$|\Delta(x)| \leq d_0, \quad |\Delta''(x)| \leq d_2, \quad |\Delta(x) - \Delta(0)| \leq \min(2d_0, d_2x^2/2). \quad (\text{AP17})$$

The last bound uses evenness. For the arithmetic sum there is also the stronger pointwise envelope

$$|\Delta(x)| \leq e^{-4x^2} \mathcal{I}(2CJ_2A(r + |x|/2)E(r + |x|/2) + J_2^2E(r + |x|/2)^2) =: \mathcal{D}(x). \quad (\text{AP18})$$

Indeed, after $y = x/2 + z$, both $|y|$ and $|x - y|$ are at most $|z| + |x|/2$. All coefficients in AP12 are positive. This proves AP18 directly for the complete residual convolution.

A.5 Every infinite tail and the Gamma endpoint

The actual source bound AP8 gives, by differentiating AP2,

$$|f(x)| \leq (95 + 175|x| + 86x^2)e^{-8x^2} \leq (183 + 174x^2)e^{-8x^2}.$$

The same bound holds for f_0 , since $C < 1/3$. Convolution of the last polynomial and retaining the Gaussian yields

$$|k(x)|, |k_0(x)| \leq Q(x)e^{-4x^2}, \quad Q(x) = 16000 + 6900x^2 + 850x^4. \quad (\text{AP19})$$

For verification the polynomial before the integer upper bounds is

$$\frac{\sqrt{\pi}}{4} [9105363/256 + (247167/16)x^2 + (7569/4)x^4];$$

$\sqrt{\pi}/4 < 4/9$ proves AP19 coefficient by coefficient.

For $L \geq 4$ put

$$T_p(L, c) = \sum_{j=0}^p \binom{p}{j} L^{p-j} j! / c^{j+1}.$$

Monotonicity of the function $v \mapsto (\log v)Q(\log v)v^{-1/2}e^{-4(\log v)^2}$ on $v \geq 64$ gives the prime-power tail bound

$$2 \sum_{n>64} \frac{\Lambda(n)}{\sqrt{n}} |k(\log n)| \leq 2 \int_4^\infty uQ(u)e^{u/2-4u^2} du < 10^{-21}. \quad (\text{AP20})$$

To check monotonicity, the logarithmic derivative in u is $1/u + Q'(u)/Q(u) - 1/2 - 8u$, with $Q'/Q \leq 4/u$. Use $\Lambda(n) \leq \log n$, then the decreasing sum-to-integral bound and $\log 64 > 4$. Expanding at $u = 4 + v$ and using $e^{-4v^2} \leq 1$ bounds the integral by

$$2e^{-62} [16000T_1(4, 63/2) + 6900T_3(4, 63/2) + 850T_5(4, 63/2)] < 100000e^{-62} < 10^{-21};$$

the bracket with its factor two is 1882788609056000/20841167403. The last inequality uses $e^3 > 20$. No prime power below the cutoff is omitted.

For either convolution, the exact Gamma tail is

$$\int_8^\infty \frac{e^{-x}k(0) - e^{-x/4}k(x/2)}{1 - e^{-x}} dx = -k(0)\log(1 - e^{-8}) - \mathcal{R}_8, \quad |\mathcal{R}_8| < 10^{-23}. \quad (\text{AP21})$$

The first term is integrated exactly. AP19 bounds the other term by

$$\frac{e^{-66}}{1 - e^{-8}} [16000T_0(8, 65/4) + 1725T_2(8, 65/4) + (425/8)T_4(8, 65/4)].$$

The bracket is $1006957513344/46411625$, its double is below 100000, and $e^{66} > 20^{22} > 10^{28}$, proving AP21.

At zero the combined Gamma integrand tends to $-3k(0)/4$. For k_0 , its integral over $[0, \epsilon]$, $\epsilon = 1/65536$, is retained within the symmetric bound

$$B_{\text{origin}} = \frac{3}{2}k_0(0)\epsilon + n_1^2\epsilon^2/8. \quad (\text{AP22})$$

In fact $1 - e^{-x} \geq x/2$, $|e^{-x} - e^{-x/4}| \leq 3x/4$ on $[0, 1]$, and $|k_0(x/2) - k_0(0)| \leq n_1^2x^2/8$. The integrand is bounded by $3k_0(0)/2 + n_1^2x/4$, which integrates to AP22.

For the full residual choose $b = 1/16$. The same combined numerator, split over $[0, b]$, $[b, 1]$, and $[1, \infty)$, proves

$$|A_\infty(k) - A_\infty(k_0)| \leq E_\Gamma := \left[\gamma_E + \log \pi + \frac{3}{2} + 4 \log(1/b) + \frac{e^{-1} + 4e^{-1/4}}{1 - e^{-1}} \right] d_0 + d_2b^2/8. \quad (\text{AP23})$$

On $[0, 1]$ the part with $\Delta(0)$ costs at most $3d_0/2$. The remaining part costs $d_2b^2/8$ below b and $4d_0 \log(1/b)$ above it. For $x \geq 1$, bound both numerator terms by d_0 times their original exponentials and use $1 - e^{-x} \geq 1 - e^{-1}$. This proves AP23.

A.6 Finite interval certificate and conclusion

The source moments are integrated with the original summands $1 \leq n \leq 16$, $0 \leq u \leq 3$, with radius 10^{-300} added for the complete omitted source. Here is the tail proof. For $u = 2 + v$, $v \geq 0$, use $e^8 > 400$ to obtain

$$u^2/32 + 9u - \pi n^2 e^{4u} \leq 145/8 - 1200n^2 - (4800n^2 - 73/8)v - (9600n^2 - 1/32)v^2 \leq -1180n^2 - v.$$

For $j \leq 4$, $(2 + v)^j \leq (2 + v)^4$, whose integral against e^{-v} is 168. The omitted part with $u \geq 2$ is therefore at most $3360 \sum n^4 e^{-1180n^2}$. Since $n^2 \geq 1 + 3(n - 1)$, the geometric fourth-moment sum at $q = e^{-3540} < 1/2$ is less than 300, giving $1008000e^{-1180} < 10^{-383}$.

For the index tail apply AP8 only to $n \geq 17$. Write $n = 17 + a$; then $n^2 \geq 289 + 35a$. For $q = e^{-105} < 1/2$,

$$\sum_{n \geq 17} n^2 e^{-3n^2} \leq e^{-867} \left[\frac{289}{1 - q} + \frac{34q}{(1 - q)^2} + \frac{q(1 + q)}{(1 - q)^3} \right] < 652e^{-867}.$$

For $j = 0, 2, 3, 4$, $20j!/3^{j+1} \leq 20/3$, so that tail is below $10000e^{-867} < 10^{-369}$. The last comparisons follow from $e^3 > 20$ and $2^{10} > 10^3$. The union of these two omitted regions contains the required omitted domain, proving the stated radius. All original summands are positive, so there is no uncontrolled cancellation in the moment-tail estimate.

The executable `certify_pairing.py` uses 100-bit ball arithmetic and `acb.integral`, with 24 subintervals for each source moment and 64 for AP5 on $[\epsilon, 8]$. Its integrands contain only entire functions and field operations; a possible division by zero returns an infinite ball. Thus no unsupported branch is declared analytic. All resulting enclosures are finite. The integration contract and error bounds are documented in the original [FLINT/Arb integration documentation](#), implemented by Fredrik Johansson and the FLINT contributors. The bounds use the resulting interval radii, not the requested tolerance alone.

The retained output gives

$$\begin{aligned} 0.06216254 < C < 0.06216255, \quad 0 < J_2 < 0.000719, \quad 0 < J_3 < 0.000120, \\ d_0 < 0.303, \quad d_2 < 175, \quad E_\Gamma < 6.1, \\ E_P := 2 \sum_{2 \leq n \leq 64} \frac{\Lambda(n)}{\sqrt{n}} \mathcal{D}(\log n) < 0.45. \end{aligned} \tag{AP24}$$

Every entry follows from finite sums of AP15 and ball integration of the explicitly displayed functions. The certificate retains the more precise intervals and checks each stated rational comparison.

Define $P_{0,64}$ using AP4 with k_0 and $n \leq 64$. AP14, the integrated Gamma interval, AP21, and AP22 give

$$56.95 < A_\infty(k_0) - P_{0,64} < 56.97. \tag{AP25}$$

Finally the exact equality

$$K = A_\infty(k_0) - P_{0,64} + [A_\infty(k) - A_\infty(k_0)] - [P_{64}(k) - P_{0,64}] - P_{>64}(k) \tag{AP26}$$

retains the entire source correction. AP20 and AP23–25 imply

$$56.95 - 6.1 - 0.45 - 10^{-21} < K < 56.97 + 6.1 + 0.45 + 10^{-21},$$

which proves AP1. No zero table, spectral truncation, presumed critical-line property, or deformed Euler product was used.

A.7 The source information carried by this sign

The original endpoint-null filter is invertible on Schwartz space:

$$h(x) = - \int_{\mathbb{R}} e^{-|x-y|/2} f(y) dy.$$

The derivative jump gives $(D^2 - 1/4)(-e^{-|x|/2}) = \delta_0$; polynomially weighted convolution bounds prove continuity on Schwartz space. The homogeneous solutions $e^{x/2}, e^{-x/2}$ have no nonzero Schwartz combination. Thus the filter retains the original source. Its two prime-boundary coordinates, in the receiver [FR15–25](#), are

$$\beta[(t_p - 1) \otimes f] = \ell_p \otimes \left(-J_2 - \frac{1089}{4} C, -\frac{\sqrt{\pi/32}}{32} e^{-8} \right). \tag{AP27}$$

The first entry follows by differentiating AP2 at zero; the second follows by integrating f and evaluating AP3 at $s = 1/2$. Both entries are negative and nonzero, while AP1 is positive. These statements concern exactly the same source and the proved maps.

Every original support label is retained: for a bounded distributive lattice L , use

$$G_L(V) = \{(v, 1_L) : v \in V\} \cup \{(0, \lambda) : \lambda \neq 1_L\}, \quad e = (0, 1_L), \quad \tau = (0, 0_L). \tag{AP28}$$

The linear source and receiver maps send $(v, 1_L)$ to $(Av, 1_L)$ and fix each lower zero label. The quadratic receiver sends a top-supported input to $(K(f), 1_L)$ and keeps the label of each zero-amplitude input. This is a specified labelled quadratic map, not a semiring homomorphism. No Boolean quotient is imposed. AP1 evaluates the amplitude of this map on the actual top-supported test.